

22 | OVERVIEW



Maximizing Net Benefits: Evaluating Climate and Health Interventions

Abstract

Climate problems are enormous, but money and resources are limited. How should leaders choose which solutions to fund? Researchers at Harvard Chan C-CHANGE, led by Lisa Robinson, use benefit-cost analysis to compare climate and health interventions systematically. Their work helps governments, funders, and advocates find the investments that do the most good.

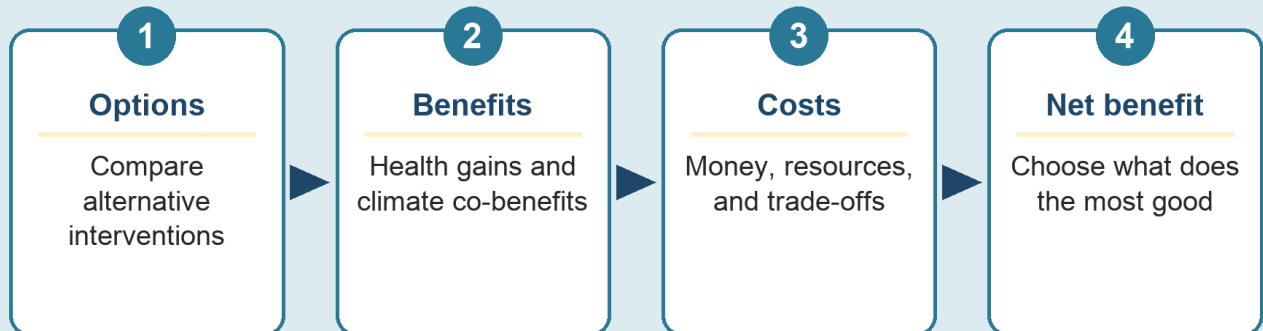
Introduction

Suppose a city has enough money for only one project: a sea wall, a network of cooling centers, or cleaner buses. Each would help, but which would help most? Policymakers face choices like this every day. Climate change makes these decisions harder, because it changes the costs and benefits of options. A project that looked sensible in the past might be in a place that will flood as sea levels rise. An intervention that once seemed minor, like reducing heat stress, may now save many more lives. Benefit-cost analysis gives decision makers a structured way to compare options and allocate resources where they will be most effective.

How Do Scientists Study This?

Benefit-cost analysis estimates all the positive and negative effects of an intervention, converts them to comparable values, and subtracts costs from benefits to find the net benefit. The C-CHANGE team works on two threads. First, they improve the methods themselves: examining the framework's strengths and limits, identifying best practices, and writing guidance, including reference case guidelines for low- and middle-income countries and guidance for the U.S. Department of Health and Human Services. They are also exploring equity-sensitive methods that consider who gains and who loses. Second, they apply these methods, reviewing research on air pollution, water and sanitation, agriculture, and nutrition to compare interventions consistently.

How benefit-cost analysis works



List the effects

Identify all the benefits and costs of an intervention, including health gains and climate effects.

Compare fairly

Use consistent methods so different options can be compared on the same scale.

Find net benefits

Subtract costs from benefits to see which choices do the most good.

What Did They Discover?

The researchers identify four ways climate change reshapes these decisions. Interventions using new technology, such as for generating energy, must be compared with alternatives. Interventions that once mattered less, such as reducing heat stress, may now yield much larger net benefits. Interventions aimed at other problems, like cutting conventional air pollution, often bring climate co-benefits that raise their value. And older analyses may need to be redone to account for extreme weather's effects on infrastructure. Their analyses inform decisions at every level, from local planning to the Paris Agreement. The team is also building a database of completed benefit-cost analyses in low- and middle-income countries to help others find cost-beneficial solutions and research gaps.

How climate change shifts the math

4

ways climate change reshapes intervention choices

2

research threads: better methods and new analyses

1

database of past analyses in lower-income countries

Benefit-cost analysis can highlight investments that deliver health benefits quickly while also slowing long-term warming, a powerful combination for policymakers.

Takeaways

Resources are limited

Leaders must choose among many climate solutions with limited money.

Climate changes the math

Rising risks can make some solutions more valuable and others less so.

Co-benefits count

Actions like cutting air pollution improve health and help the climate.

Key Terms

Benefit-cost analysis

A method for comparing an option's total benefits with its total costs to judge whether it is worthwhile.

Net benefit

The total benefits of an action minus its total costs.

Co-benefit

An additional positive effect of an action beyond its main goal, such as health gains from cutting emissions.

Equity-sensitive analysis

An approach that considers how benefits and costs are distributed across different groups of people.

Low- and middle-income countries

Nations with lower average incomes, where many cost-effective health interventions are especially needed.

Match-Up: draw a line from each term to its clue

A Benefit-cost analysis

B Net benefit

C Co-benefit

D Equity-sensitive analysis

E Low- and middle-income countries

1 An additional positive effect of an action beyond its main goal

2 An approach that considers how benefits and costs are distributed across different...

3 A method for comparing an option's total benefits with its total costs...

4 Nations with lower average incomes, where many cost-effective health interventions are especially...

5 The total benefits of an action minus its total costs

Answer key: A-3 · B-5 · C-1 · D-2 · E-4

Activity

The City Budget Challenge

Students compare three climate and health projects using a simple benefit-cost table.

Materials: Project cards: tree planting, cooling centers, electric buses; Benefit-cost worksheet; Calculator

1. For each project, list at least three benefits and two costs. Include health benefits and climate co-benefits.
2. Give each benefit and cost a score from 1 to 10, then calculate a net benefit score.
3. Rank the projects. Which has the largest net benefit?
4. Now add a climate twist: the neighborhood will face many more heat waves. Re-score and see whether your ranking changes.
5. Discuss: who benefits most from each project? Would an equity-sensitive analysis change your choice?

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Evaluating climate and health interventions to maximize net benefits.

<https://hsph.harvard.edu/research/climate-health-c-change/research/evaluating-climate-and-health-interventions-to-maximize-benefits/>